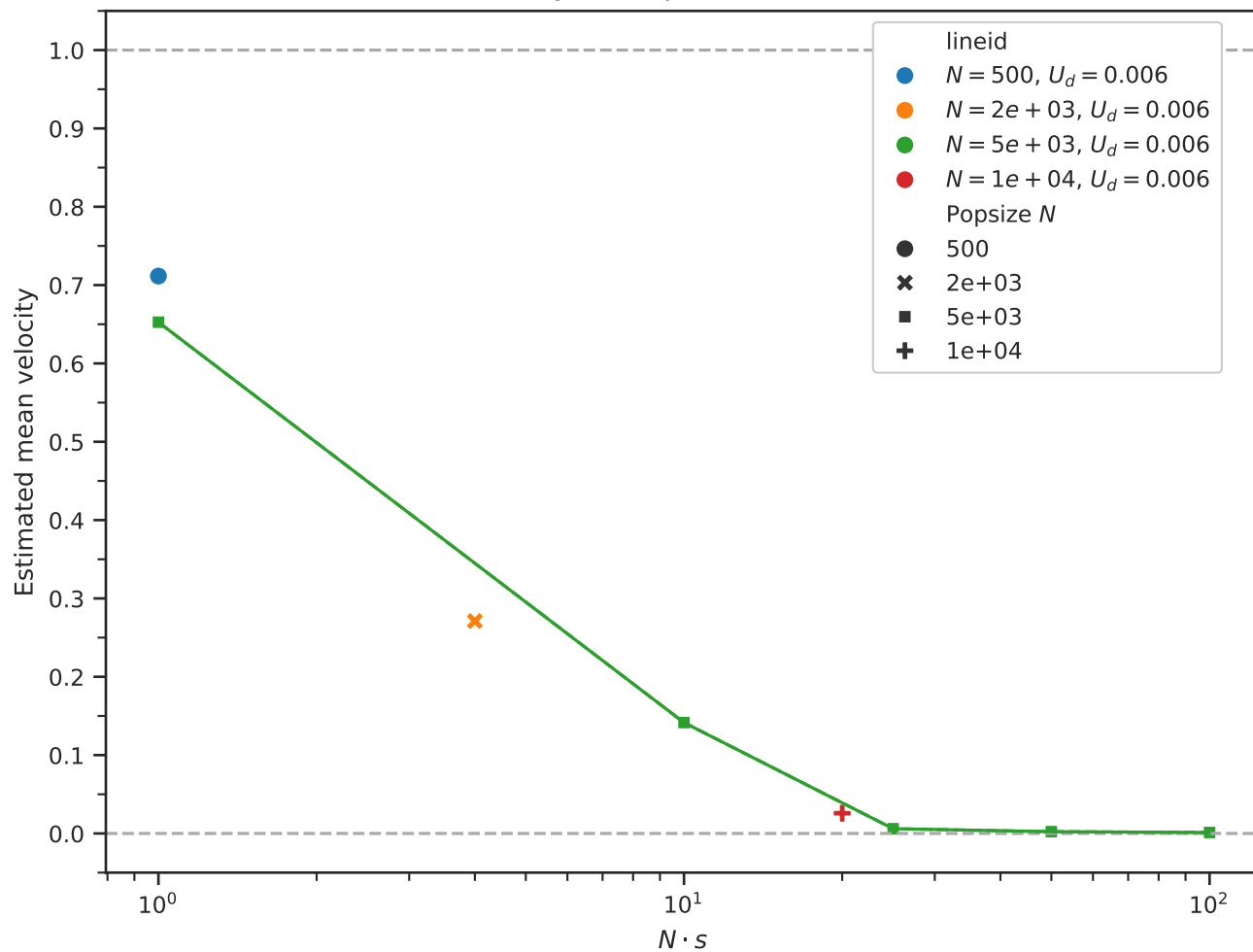
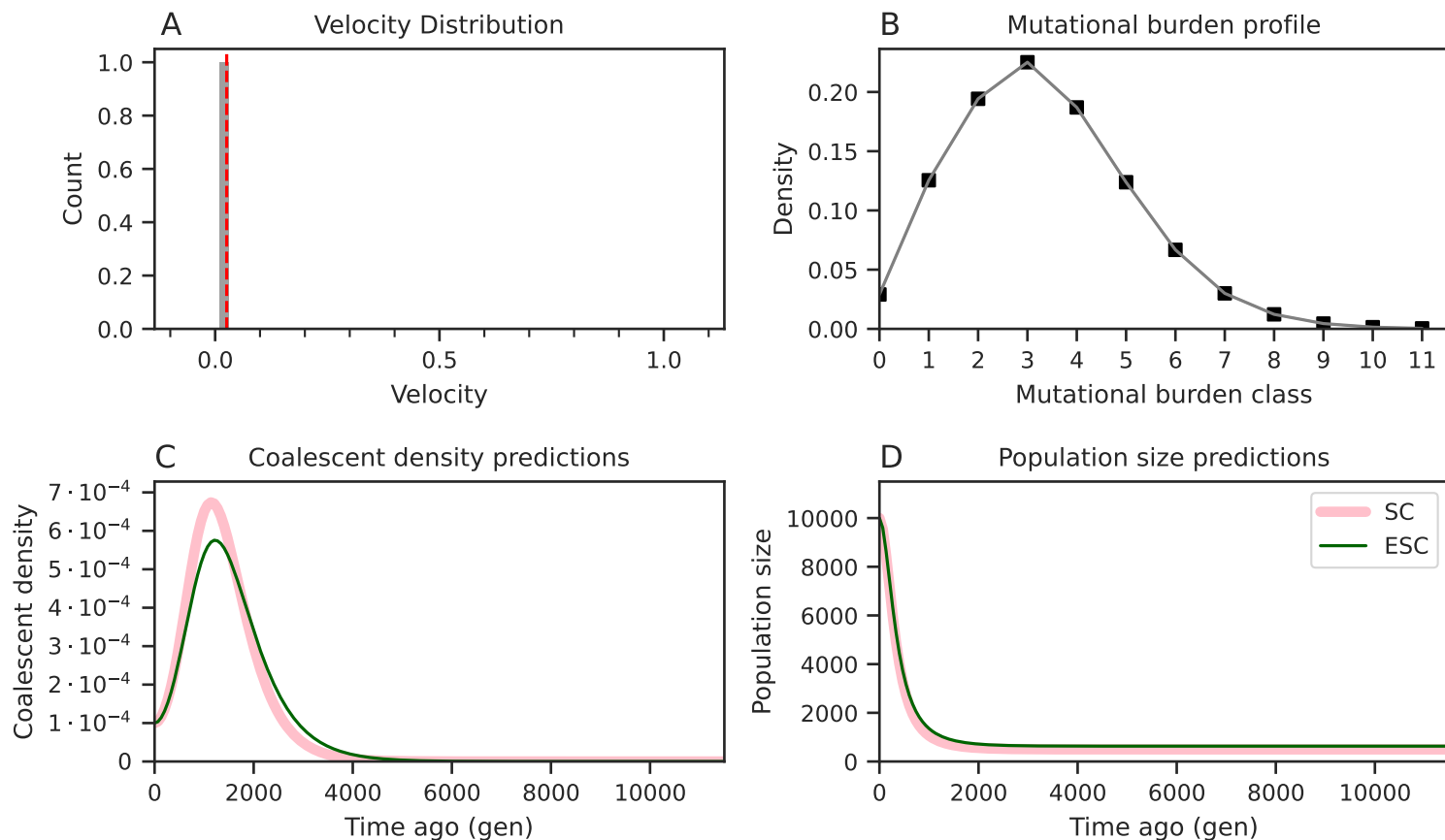


Mean velocity across parameter combinations

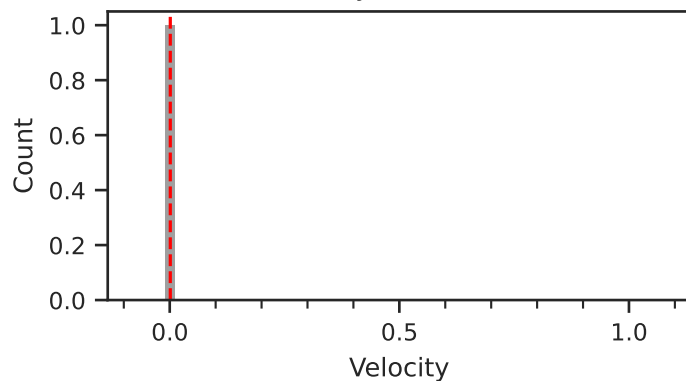


$N = 1e + 04$, $s = 0.002$, $U_d = 0.006$, $\phi = 0.996$
Mean Velocity= 0.0258 ± 0.0000 , $n = 1$ simulations

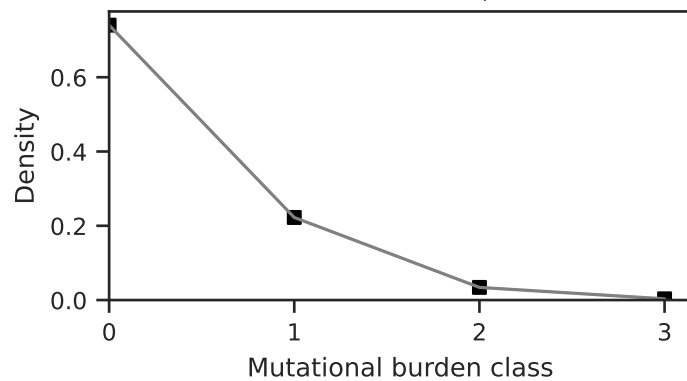


$N = 5e + 03$, $s = 0.02$, $U_d = 0.006$, $\phi = 74.1$
Mean Velocity= 0.0012 ± 0.0000 , $n = 1$ simulations

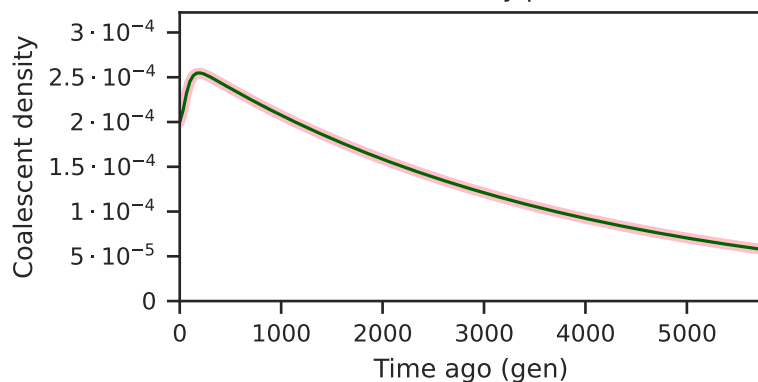
A Velocity Distribution



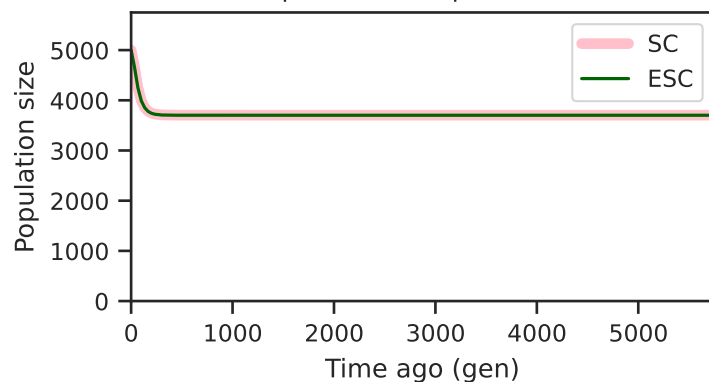
B Mutational burden profile



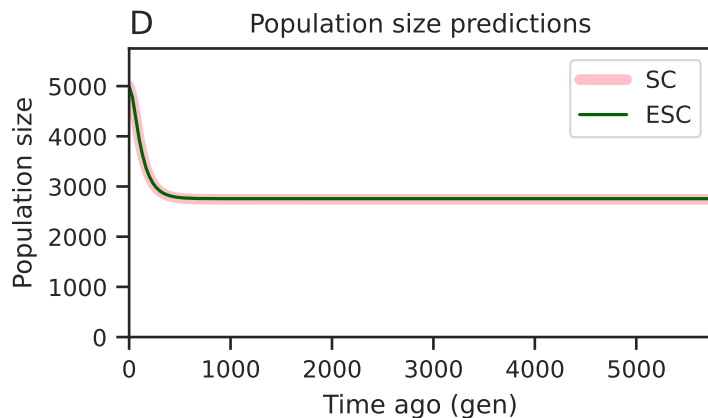
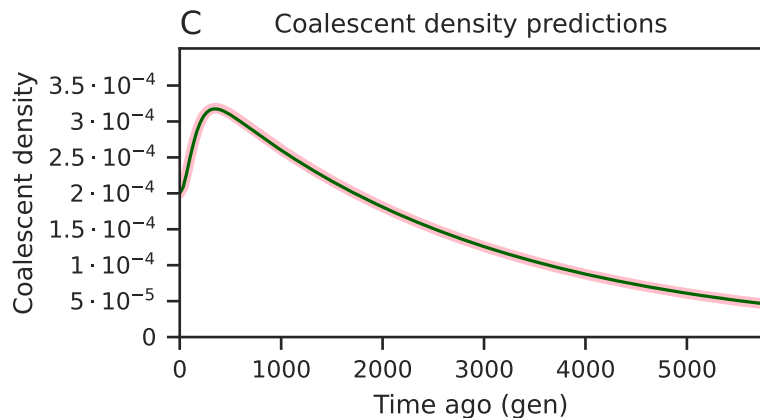
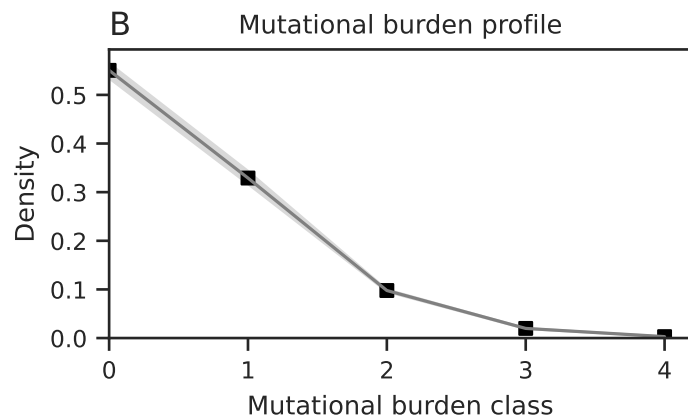
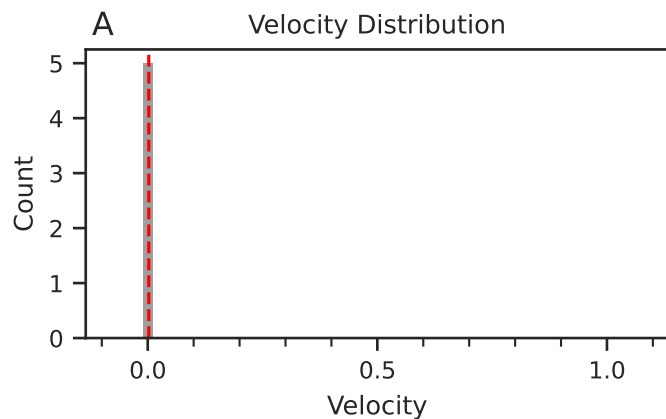
C Coalescent density predictions



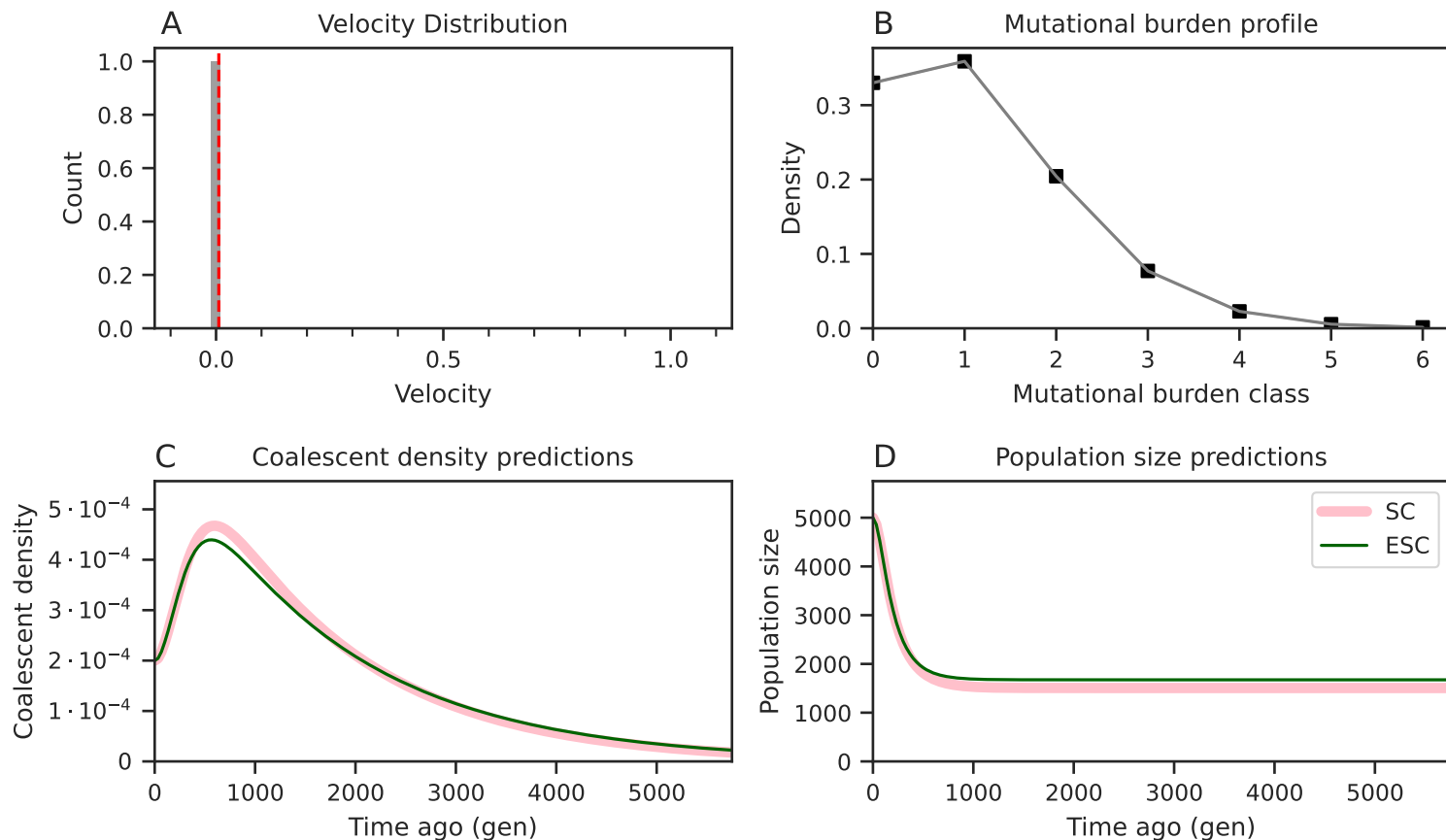
D Population size predictions



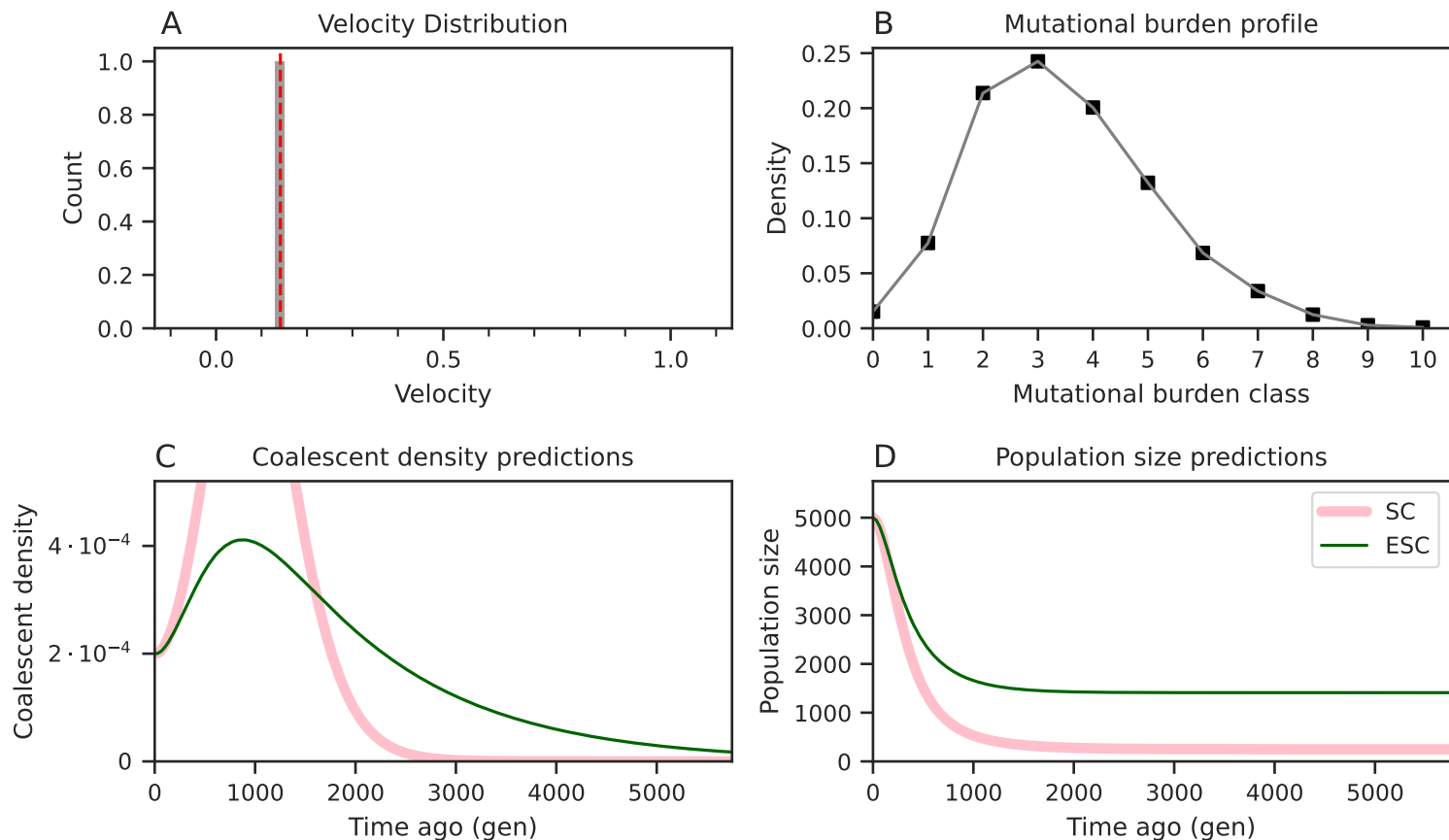
$N = 5e + 03$, $s = 0.01$, $U_d = 0.006$, $\phi = 27.4$
Mean Velocity= 0.0023 ± 0.0014 , $n = 5$ simulations



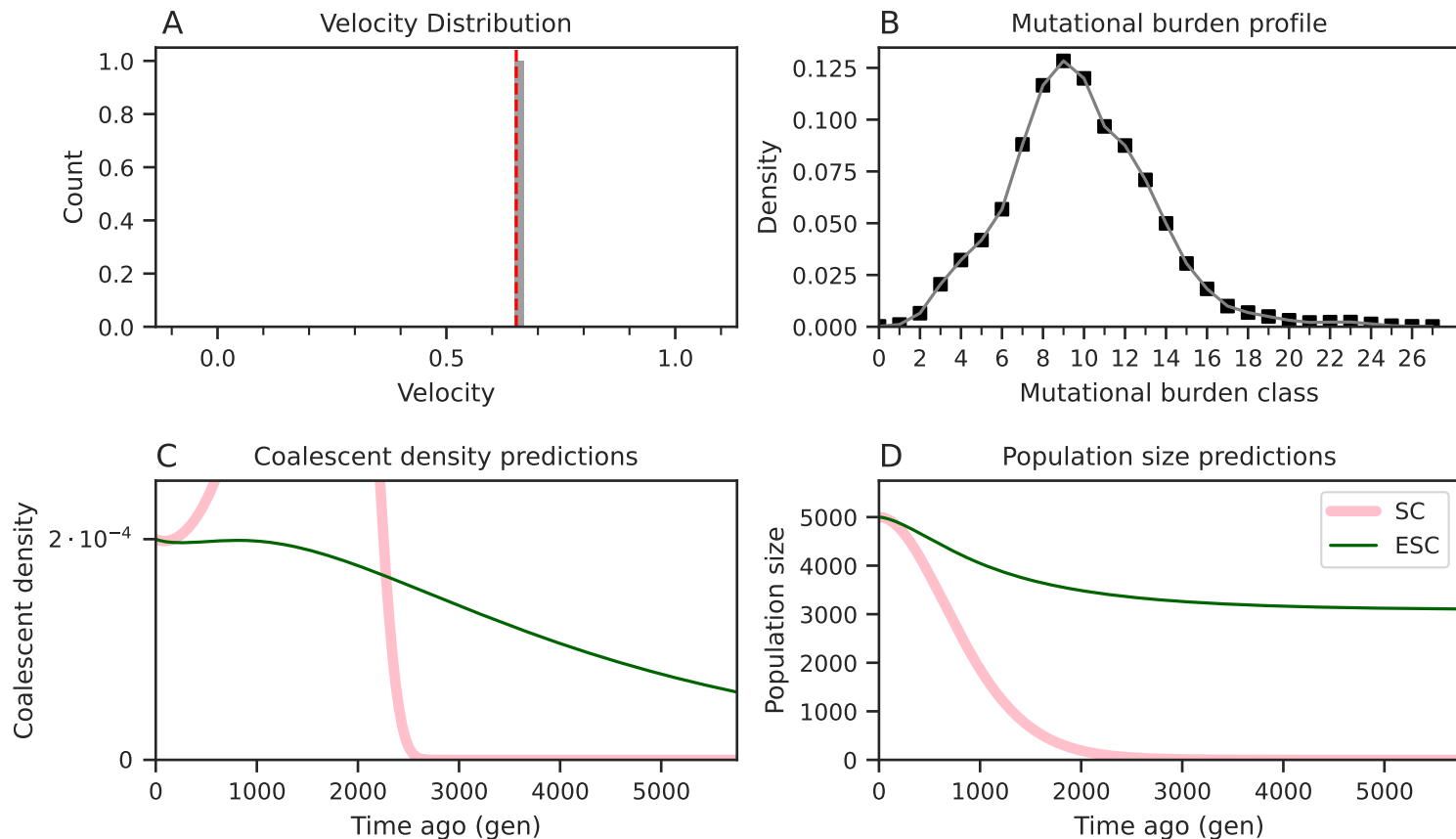
$N = 5e + 03$, $s = 0.005$, $U_d = 0.006$, $\phi = 7.53$
Mean Velocity= 0.0060 ± 0.0000 , $n = 1$ simulations



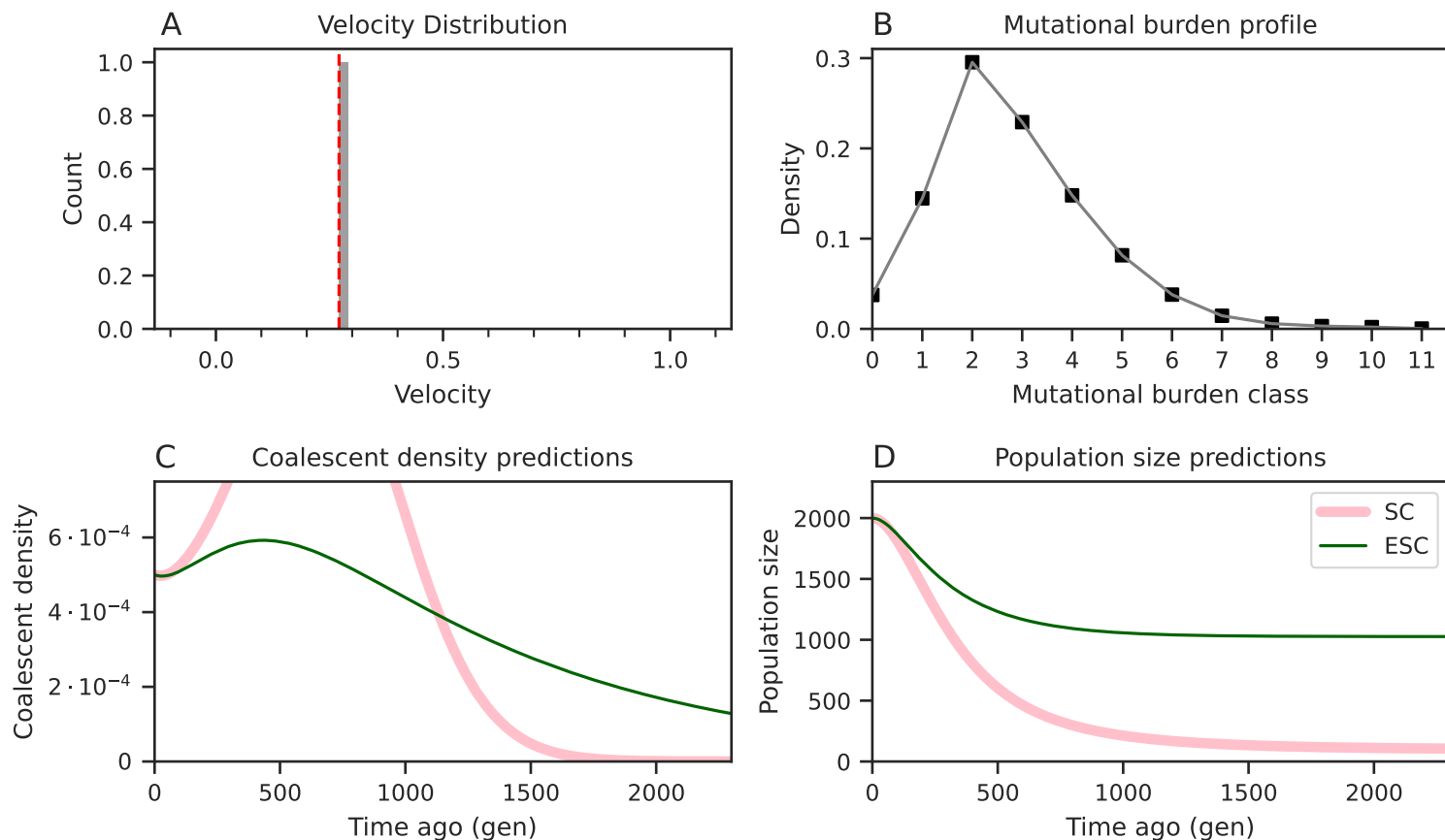
$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\phi = 0.498$
Mean Velocity= 0.1415 ± 0.0000 , $n = 1$ simulations



$N = 5e + 03$, $s = 2 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 9.36 \cdot 10^{-14}$
Mean Velocity= 0.6526 ± 0.0000 , $n = 1$ simulations



$N = 2e + 03$, $s = 0.002$, $U_d = 0.006$, $\phi = 0.199$
Mean Velocity= 0.2711 ± 0.0000 , $n = 1$ simulations



$N = 500$, $s = 0.002$, $U_d = 0.006$, $\phi = 0.0498$
Mean Velocity= 0.7115 ± 0.0000 , $n = 1$ simulations

